12. How will you avoid the growing of microbes in nutrient medium during culture process? What are the techniques used to remove the microbes?
13. Write the various steps involved in cell suspension culture.
14. What do you mean Embryoids? Write its application.
15. Give the examples for micro propagation performed plants .
16. Explain the basic concepts involved in plant tissue culture.
17. Based on the material used, how will you classify the culture technology? Explain it.
18. Give an account on Cryopreservation.
19. What do you know about Germplasm conservation. Describe it.
20. Write the protocol for artificial seed preparation.

Glossary

Aseptic condition: Preparation of materials free from microbes in *in vitro* cultures.

Cell Culture: Growing of cells *in vitro*, including the culture of single cells or small aggregates of cells in a liquid medium.

Chemically defined medium: A nutritive medium used for culturing cells or tissue; each chemical of this medium is known and defined;

Cybrid: Cytoplasmic hybrid obtained by the fusion of cytoplasm of cells of different parental sources; a term applied to the fusion of cytoplasms of two different protoplasts;

Organogenesis: The process of initiation and development of shoot or root though *in vitro* culture particularly from callus

APPENDIX

Composition of MS(Murashige and Skoog) Medium

Macronutrients:	
Ammonium nitrate (NH_4NO_3)	1650.0 mg/l
Potassium nitrate (KNO3)	1900.0 mg/l
Calcium chloride (CaCl2 2H2O)	440.0 mg/l
Magnesium sulphate (MgSO4 6H2O)	370.0 mg/l
Potassium dihydrogen phosphate (KH2PO4)	170.0 mg/l
Micronutrients:	
Manganese sulphate (MnSO4 4H2O)	22.3 mg/l
Zinc sulphate (ZnSO4 4H2O)	8.6 mg/l
Boric acid (H3BO3)	6.2 mg/l
Potassium iodide (KI)	0.83 mg/l
Minor nutrient:	
Sodium molybdate (Na2 MO4 2H2O)	0.250 mg/l
Cupric sulphate (CuSO4 5H2O)	0.025 mg/l
Cobaltous chloride (CoCl2 6H2O)	0.025 mg/l
Iron stock	
Na EDTA	37.25 mg/l
Ferrous Sulphate (FeSO4 7H2O)	27.85 mg/l
Vitamins	
Glycine	2.0 mg/l
Nicotinic acid	0.5 mg/l
Pyridoxin HCl	0.5 mg/l
Thaiamine HCl	0.1 mg/l
Growth Hormones	
IAA	1.30 mg/l
Kinetin	0.4–10.0 mg/l
Myo-inositol	100.0 mg/l
Sucrose	30.0 g/l
Solidifying Agent	
Agar	8.0 g/l

TN_GOVT_BOTANY_XII_PAGES_110-122 CH 05.indd 121 23-12-2022 10:59:59